

Text, Screenshots, or Both?

Representation Value and Task Routing in Web Agents

Anonymous ACL submission

Abstract

A web agent can read a page as text, look at a screenshot, or do both, and deployments usually fix one choice for every task. We measure that choice across three sites of VisualWebArena and WebArena with four vision-language backbones, using text-only variants as controls and rerunning identical conditions to see which differences survive. Three findings separate questions that are usually merged. First, no observation choice is best everywhere, and the screenshot has a real, reproducible value that is predictable in advance on one site: a zero-cost rule over the task text picks out classifieds tasks on which adding the screenshot alone raises success by 25–30 points for the two strongest backbones (11 for a 4B model), and the gap reappears on a full rerun. Second, which representation suits which task is a genuine property of the task, beyond its difficulty, yet a single run measures it with reliability of only 0.31–0.50. Third, routers that choose a representation per task, with their operating point fixed on training data, do not convert this signal into a gain over the best fixed choices and their random mixtures. We describe the rerun-controlled protocol that keeps these three quantities apart.

1 Introduction

An agent that operates a website has to perceive it, and the two standard ways are complementary in principle: the page’s text (an accessibility tree) is precise and cheap, a screenshot shows what the text omits. A third option fuses them, overlaying numbered marks on the screenshot and listing the marked elements as text (Yang et al., 2023; Koh et al., 2024). Practitioners typically pick one of the three for a whole deployment. The question this paper asks is the one that choice raises: *is the choice worth making per task, and can it be made?*

Answering it requires keeping apart three quantities that a single leaderboard number merges:

1. **Representation value:** does the screenshot change success, holding everything else fixed?
2. **Preference reliability:** when one representation beats another on a task, is that a stable property of the task, or a property of one run?
3. **Routing gain:** can a policy that chooses per task, using only what is known before it runs, beat the fixed choices?

A large hindsight ceiling — pick, for each task, whichever representation happened to succeed — answers none of them, because it rewards run-to-run luck as much as structure. We therefore rerun identical conditions and read every comparison against what a rerun alone changes.

Findings. (i) No choice wins everywhere: the fused mode leads on classifieds, text leads on WebArena reddit, and the order reverses across backbones (§3). (ii) The screenshot has a specific value that can be isolated and predicted: with the text payload and prompt held fixed, adding the screenshot raises success on the classifieds tasks a zero-cost intent rule flags by 25–30 points for the two strongest backbones (11 for a 4B model), against 5–10 on the other tasks; the effect reproduces on a full rerun and is largest for the strongest backbone. On reddit the same rule finds nothing (§4). (iii) Per-task preferences between representations exist beyond task difficulty — a no-interaction null conditioned on difficulty is rejected in every fully rerun cell — but a single run measures them with reliability 0.31–0.50; two to three runs per task are needed to pass 0.5 (§5). (iv) Routers that fix their operating point on training data do not beat the fixed representations and their random mixtures; the room that exists above that frontier comes from task difficulty, not from representation fit, and is not captured without looking at test outcomes (§6).

Contributions. A rerun-controlled measurement of text, screenshot and fused observation across

three sites and four backbones, with text-only ablations that isolate the screenshot; a matched estimate of the screenshot’s value and an ex-ante rule that predicts it on one site; a test of whether per-task representation preferences are real, with a decision study for how many runs make them learnable; and a deployable-policy comparison against fixed choices *and their random mixtures* — the baseline of model-routing benchmarks, applied here to observation choice — which is the comparison a routing claim has to win (§7).

2 Setup

Sites and tasks. We use three sites of VisualWebArena (Koh et al., 2024) — classifieds (224 scored tasks), reddit (203) and shopping (432) — and the reddit site of WebArena (Zhou et al., 2024) (104 tasks, those all observation modes ran). VWA-reddit and WA-reddit are different task sets on the same application, so they are not two independent benchmarks. Scored sets are the collected tasks minus protocol exclusions written as rules over task configuration (Appendix A).

Backbones. B0 = Qwen3-VL-235B (API-served), B1 = Qwen3-VL-4B and B2 = Gemma-3-4B (both served locally), and B5 = GPT-5.6 (API, classifieds only). A *cell* is one site $\times$ backbone pair; the eight cells with every observation mode are VWA-classifieds and VWA-reddit $\times$ {B0, B1, B2} and WA-reddit $\times$ {B0, B1}. B5-classifieds and the two shopping cells extend them (§4, §6) and are never pooled with the eight.

One plain agent. One model call per step emits a thought, an action and its arguments; the prompt holds the instruction, the mode’s system prompt, the last eight steps and the current observation. There is no planner, memory, reflection or retry policy, so a difference between conditions is a difference of observation, not of scaffold. Episodes stop at 30 steps.

Observation modes. The deployment question has three answers: **DOM** (the page’s accessibility tree as text), **Vision** (the raw screenshot, with an empty text payload) and **SoM** (a set-of-marks screenshot plus the list of marked elements as text; Yang et al., 2023). Three further modes carry no screenshot and serve as controls; the one used throughout is **P-SoM**, which is SoM with the screenshot removed — same marks, same prompt — so SoM–P-SoM isolates the image (Appendix B

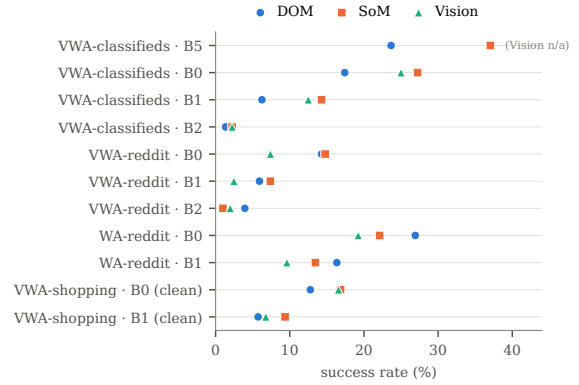


Figure 1: Success rate of the three deployment choices per cell. Shopping cells drop the tasks touched by a known harness defect in any mode (§2, Appendix D); B5’s Vision run is excluded for a coordinate-contract defect. Source: routing_three_arm, routing_extension_cells.

lists all six). “Text” modes withhold the *page* screenshot only: a reference image that is part of the task (“find this item”) is given in every mode.

Reruns. We reran identical conditions, 24 registered same-condition pairs in all; in three cells — VWA-classifieds B0, VWA-reddit B0 and WA-reddit B1 — every mode was run twice. A rerun uses the same tasks, code and site reset, so what it changes is what one run of the agent does not determine.

Scoring and cost. The evaluators are binary. Cost is the per-episode billed cost: an API invoice for B0 and B5, a token-priced electricity estimate for B1 and B2 (per-cell values in Table 2). The two kinds are ordered within a cell, never pooled across; §6 repeats the routing analysis under GPU-time and wall-clock cost.

3 No observation choice is best everywhere

Figure 1 shows the three deployment choices side by side. On classifieds the fused mode is the best of the three for B0, B1 and GPT-5.6 (B2 solves too few tasks to order the modes), and the margin over text is not small: SoM–DOM is +9.82 pp [+3.57, +16.07] for B0 and +8.04 pp [+3.57, +12.95] for B1 (paired task bootstrap). On WebArena reddit the order flips and text leads (SoM–DOM –4.81 pp for B0 and –2.88 pp for B1, both intervals crossing zero), and on VWA-reddit the weakest backbone does significantly worse with the fused mode

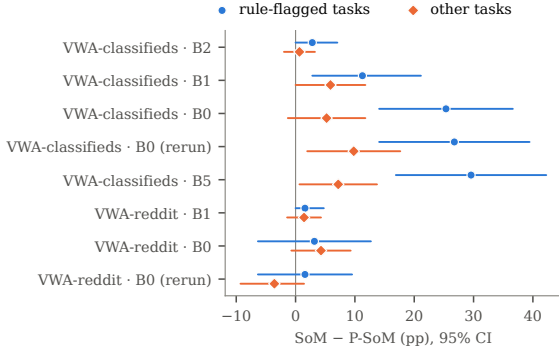


Figure 2: Value of the screenshot alone (SoM–P-SoM: same marks, same prompt) on the tasks a zero-cost intent rule flags, against the other tasks. Classifieds is out-of-sample for the rule. “Rerun” is a full repeat of both modes. VWA-reddit B2 is omitted: no mode solves a flagged task. Source: visual_intent_routing.

(−2.96 pp [−5.91, −0.49]). On shopping the fused mode is best or tied with Vision (Figure 1).

Two cautions keep this descriptive. Cells on the same site share their tasks, so the eight are not eight independent replications. And the screenshot-free modes are four, not one: a maximum over four text variants against a single screenshot mode would be a selection effect, so the comparison here takes one representative per deployment class, as practitioners do.

What this section establishes is only that the choice matters and depends on the workload. Whether it can be made per task is the subject of the rest of the paper.

4 The screenshot has a value that can be isolated and predicted

A matched contrast. Comparing Vision with DOM changes two things at once — the image and the text the agent receives. SoM and P-SoM differ only in the annotated screenshot, so their difference is the value of the image alone. Chi et al. (2024) already compared screenshot-plus-text with text only on a subset of VisualWebArena tasks for one backbone (50.0% against 46.3%, without set-of-marks); we hold the set-of-marks text and prompt fixed, use four backbones and full task sets, rerun the contrast, and ask *where* the image pays off.

An ex-ante rule. A regular expression over the task intent flags tasks that ask about an image, a photo, a colour or a count of things shown, excluding tasks that already carry a reference image

(Appendix C). It reads only the task configuration: no model call and no episode. It was written for reddit, so its classifieds hits were never inspected and classifieds is out-of-sample.

Result. On the 71 classifieds tasks it flags, the screenshot raises success by +25.35 pp [+14.08, +36.62] for B0, against +5.23 pp on the other 153 tasks (Figure 2). The effect reproduces on a full rerun of both modes (+26.76 pp [+14.08, +39.44], from 23 against 4 successes), and it grows with capability: +2.82 pp for B2, +11.27 pp for B1, +25.35 pp for B0 and +29.58 pp [+16.90, +42.25] for GPT-5.6. On VWA-reddit the same rule flags 63 tasks and separates nothing: +3.17 pp on flagged tasks against +4.29 pp elsewhere for B0, +1.59 pp against −3.57 pp on its rerun. On shopping the flagged tasks carry at most five successes per mode, too few to read, and on WebArena the rule flags only 5 of 104 tasks.

Reading. Where the screenshot pays off on classifieds is predictable before anything runs, and it pays off more for stronger models; on reddit it is not. The screenshot’s value is therefore real and partly foreseeable, but site-specific: it is a statement about a kind of task, not a general law. It is also not a router. On these tasks the fused mode already contains the screenshot, so switching between DOM and Vision by this rule cannot beat simply using SoM; what remains open is whether *individual* tasks have stable preferences beyond such task types, which is the question of §5.

5 Per-task preferences are real, but one run cannot measure them

§4 found structure at the level of task *types*. Routing per task needs more: that an individual task prefers one representation over another, beyond being easy or hard, and that this preference is stable enough to learn from.

A null without task-by-mode structure. In the three cells where every mode was run twice, let $Y_{imr} \in \{0, 1\}$ be the outcome of task i under mode m in run r . Under

$$\text{logit } \Pr(Y_{imr} = 1) = a_i + b_{mr},$$

every task has its own difficulty and every (mode, run) column its own easiness, but there is no persistent task-by-mode interaction. This is the Rasch structure that item-response routers use to model

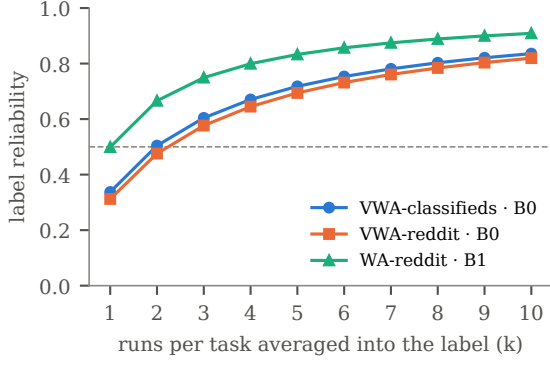


Figure 3: Decision study for the three deployment modes: reliability of a task-by-mode label averaged over k runs, $\sigma_{\text{int}}^2/(\sigma_{\text{int}}^2 + \sigma_{\text{err}}^2/k)$. A single run gives 0.31–0.50. Source: routing_three_arm.

difficulty (Song et al., 2025); here it is the null, and the question is what lies beyond it. The row and column sums are sufficient for this model, so conditional on them every 0/1 matrix with those margins is equally likely; we sample that null exactly with curveball swaps, fitting nothing. The statistic is the reproducible part of the interaction, $\hat{\sigma}_{\text{int}}^2 = (\text{MS}_{\text{int}} - \text{MS}_{\text{err}})/2$ from the two-way task $\times$ mode ANOVA with two runs: the run-to-run covariance of each task’s mode profile.

Result. Over the three deployment modes the interaction exceeds the null in all three cells (classifieds B0 $\hat{\sigma}_{\text{int}}^2 = 0.033$ against a null 95th percentile of 0.015; reddit B0 0.018 against 0.009; WebArena-reddit B1 0.030 against 0.014; $p \leq 0.002$ each, Holm across cells). Over all six modes the same holds (Appendix E). Preferences between representations are therefore a property of tasks, not only of their difficulty.

But a single run measures them poorly. The same decomposition gives the reliability of a label built from k runs (Figure 3). From one run it is 0.34, 0.31 and 0.50 in the three cells; two to three runs per task are needed to pass 0.5. Over six modes it is lower still, 0.14–0.46, and up to six runs are needed. The preferences also persist where they can be observed directly: taking, for each task, the cheapest mode that succeeded on one run and scoring that choice on the rerun beats the fixed-mode frontier by 2.4–10.5 points on reddit B0 and WebArena-reddit B1. That choice uses the task’s own label, which no deployment has, so it measures persistence, not an attainable policy.

The routing target thus exists, and a training set

built from single runs labels it with a reliability between a third and a half.

6 Deployable routers do not convert the signal

The comparison a router must win. A router that sends some tasks to an expensive mode and others to a cheap one trades success for cost. Beating every fixed mode on both axes is too strict, but beating the fixed modes one at a time is too lenient: sending a random share of tasks to each of two modes is also a deployable policy. We therefore compare against the *frontier of the fixed modes and all their random mixtures* — the upper concave envelope of the fixed points in the (cost, success) plane, the non-decreasing convex hull that model-routing benchmarks use as a baseline (Hu et al., 2024; Jitkrittum et al., 2025) — and report how far above it a router reaches (Figure 4a; definitions in Appendix H).

Routers. Two constructions use 18 features available before a task runs: 14 intent-keyword indicators, the intent length, and three statistics of the first page. *Per-arm heads* fit one logistic success model per mode and send a task to the cheapest mode whose predicted success clears a threshold. *Triage* predicts whether any mode can solve the task, sends predicted-hopeless tasks to the cheapest mode and the rest to the best one. All predictions are out-of-fold (task-held-out five-fold).

Room above the frontier exists, and it is difficult. Sweeping each router’s threshold traces a curve. Over the three deployment modes, the triage curve pooled over the eight cells rises 1.69 points above the frontier at low budget (Figure 4b), which a label-shuffle null that refits the whole curve does not produce ($p = 0.001$; Table 1). The per-arm heads reach 0.86 points ($p = 0.072$). Triage does not choose a representation for a task; it decides which tasks are not worth the expensive mode. The room that exists is the difficulty lever, not representation fit.

A deployable router does not capture it. A curve’s best point is chosen after seeing test outcomes. When the operating point is instead chosen on training folds only, the pooled gain is +0.23 points for per-arm heads and +0.59 for triage, and neither interval excludes zero (Table 1). Over all six modes the deployable per-arm router sits significantly *below* the frontier (−0.38; 95% upper

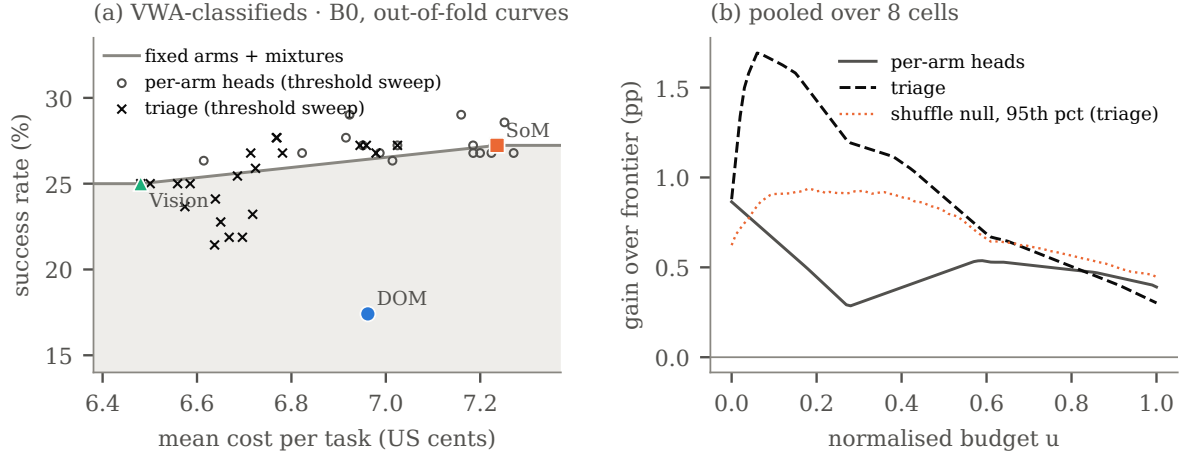


Figure 4: (a) One cell’s cost–success plane: the three deployment arms, the frontier of the arms and their random mixtures (shaded: attainable without routing), and the out-of-fold points of the two routers as their threshold is swept. (b) Gain of the frontier when each router’s curve is added, averaged over the eight cells on a budget normalised so that the cheapest arm is $u = 0$ and the dearest $u = 1$, against the 95th percentile of a label-shuffle null that refits the whole curve. Both panels choose the best point after seeing test outcomes; Table 1 gives the deployable version. Source: routing_three_arm.

cells	policy	curve max (p)	deployable [5%, 95%]
3 arms, 8 cells pooled	per-arm heads	+0.86 (0.072)	+0.23 [−2.20, +0.22]
	triage	+1.69 (0.001)	+0.59 [−1.53, +1.23]
6 arms, 8 cells pooled	per-arm heads	+0.67 (0.252)	−0.38 [−3.20, −0.14]
	triage	+1.24 (0.043)	−0.39 [−2.42, +0.81]
GPT-5.6, classifieds (5 arms)	per-arm heads	+0.45 (0.768)	−2.45 [−8.61, +4.24]
	triage	+3.86 (0.127)	−1.63 [−6.70, +5.74]
Shopping B0 (3 arms)	per-arm heads	+1.14 (0.062)	−0.18 [−2.45, +2.02]
	triage	+0.20 (0.496)	−0.87 [−2.63, +1.37]
Shopping B1 (6 arms)	per-arm heads	+1.11 (0.143)	+0.56 [−4.69, +0.88]
	triage	+1.63 (0.158)	+1.19 [−4.10, +1.57]

Table 1: Routing gain over the fixed arms and their random mixtures, in SR points. *Curve max*: the best point of an out-of-fold curve, chosen after seeing test outcomes, with its label-shuffle p (pooled rows: max over a normalised budget). *Deployable*: the operating point chosen on training folds only, task bootstrap. Sources: routing_three_arm, representation_routing_frontier, routing_gain_upper_bounds, routing_extension_cells.

bound -0.14). The pattern holds for the strongest backbone: for GPT-5.6 on classifieds, where the screenshot’s value was largest (§4), the deployable routers lose 1.6–2.5 points; on both shopping cells they stay within 1.2 points of the frontier.

Robustness. Holding tasks of the same template out together lowers the six-mode per-arm router’s mean deployable gain from -0.38 to -1.60 points, and fitting on one run and scoring on its rerun gives point estimates of both signs whose intervals all include zero (Appendix F). Repricing cost as GPU time or wall-clock time leaves the frontier result unchanged, but changes *which* single cell’s triage curve passes a per-cell Holm correction, so no single-cell result is quoted here. One of the

three-mode intervals (per-arm heads, pooled) has an upper end below its point estimate, a sign that the bootstrap of a policy with a selected threshold is biased; we read these intervals as magnitudes, not as tests.

7 A protocol for multi-representation claims

The three quantities of §1 came apart because each was measured against its own control. We suggest the same four steps for any claim that choosing among several agent configurations per task would pay.

1. **Report reruns per arm and per cell**, not one flip rate or one resolution threshold for

a whole study: in our data a rerun changes between 0 and 14% of outcomes depending on arm and cell (Appendix E).

2. **Isolate the component of interest with a matched ablation** (here SoM–P-SoM, which removes only the screenshot) before attributing a difference to it.
3. **Test for per-task structure against a null that conditions on difficulty**, and use the decision study to say how many runs per task a training label needs. A single-run oracle is not a ceiling a router can reach.
4. **Score a deployable policy, operating point chosen on training data, against the fixed arms and their random mixtures**, never only against the best single arm or a hindsight curve.

Applied here, the protocol says the screenshot is worth paying for on a predictable class of classified tasks, that per-task representation preferences are real but need two to three runs per task to be learnable, and that with single-run labels and pre-flight features a per-task representation router does not yet beat choosing one representation, or a fixed mixture, for the whole workload.

8 Related work

How web agents observe the page. Agents have read HTML or accessibility trees (Deng et al., 2023; Yang et al., 2025), screenshots grounded to elements (Zheng et al., 2024; He et al., 2024; Gou et al., 2025), or set-of-marks overlays that fuse the two (Yang et al., 2023; Koh et al., 2024). A parallel line shrinks or reshapes the text observation (Kerboua et al., 2025; Schiepaniski and Piël, 2025; Enomoto et al., 2026), and some systems combine several grounding routes (Li et al., 2026a). Chi et al. (2024) compare screenshot-plus-text with text only on a VisualWebArena subset. Closer to adaptivity, Li and Shi (2026) gate visual observation components during execution of computer-use tasks, and Liu et al. (2026) route each action of a computer-use agent among vision-language models; neither chooses between text and screenshot per task before it runs. These works compare observation designs at the level of a benchmark score or adapt them online; we ask whether the choice should be made per task in advance, and isolate the screenshot with a matched ablation rather than comparing whole pipelines.

Routing and cascades. Choosing among models per query to trade quality for cost is well studied for language models (Chen et al., 2024; Ong et al., 2025; Ding et al., 2024; Gupta et al., 2024; Peale et al., 2026); see Moslem and Kelleher (2026) for a survey. For web agents, Li et al. (2025) learn query-specific routing for cost-sensitive agents, and Hajimiri et al. (2026) judge agent components at a matched budget. Early exit (Lu et al., 2025) stops agents that are unlikely to succeed — the difficulty lever that, in our data, is the only part of routing that leaves room above the fixed frontier. Routing benchmarks compare routers against random mixtures of fixed models (Hu et al., 2024; Jitkrittum et al., 2025), and under unified conditions several learned routers fail to beat a simple baseline (Li et al., 2026b); Lee (2026) finds that assigning one model per task type in advance recovers most of the oracle routing gain on question answering, and that reruns rescore 5% of model–question pairs. Item-response routers model query difficulty and model ability (Song et al., 2025). We route over observation representations rather than models, take the mixture baseline from this literature, and add what it rarely has: reruns of identical agent conditions and a test of per-task structure beyond difficulty.

Reliability of agent evaluation. Repeated trials of the same agent disagree, which τ -bench measures with pass^k (Yao et al., 2025); inference itself is not deterministic even at fixed settings (He, 2025; Yuan et al., 2025), prompts change outcomes in ways unrelated to content (Sclar et al., 2024), and agent benchmarks carry evaluator and task defects (Hattami et al., 2025; Bhat et al., 2026). Our reruns quantify the first for web agents at the task level and show what it does to per-task routing labels; the harness defects we found are listed in Appendix D.

Limitations

Coverage of reruns. Only three cells have every mode run twice, two with the same backbone; the preference test and decision study (§5) rest on them. Five of the eleven cells have no rerun of any mode.

Breadth. VWA-reddit and WA-reddit are task sets on the same application, so the study covers three applications, not four benchmarks. GPT-5.6 was run on classifieds only, and its Vision run is excluded for a coordinate-contract defect; a parser defect also let it take actions beyond the 30-step budget in every mode (Appendix D). The ex-ante

rule of §4 works on classifieds only.

Shopping filtering. Dropping tasks touched by the search-box defect keeps the comparison between modes clean, but the dropped tasks are mostly search tasks, so the filtered cells read a different task mix; unfiltered results are reported alongside.

What the routing result does not show. The routers use 18 pre-flight features and logistic models; richer features, language-model routers or multiple runs per training task may do better. The result is that, with single-run labels and these features, the routers do not beat the fixed choices and their mixtures — not that per-task routing is impossible. Step-level switching is out of scope.

Exploratory analyses. The preference test, frontier comparisons and bounds were designed after the runs were collected. Each family states its multiplicity correction; the bootstrap intervals for policies with a selected threshold are read as magnitudes, not tests.

Cost. API invoices and electricity estimates are different quantities and are never pooled; local cost depends on the pricing basis, which is why §6 repeats the analysis under GPU time and wall-clock time.

Ethics statement

All episodes ran against self-hosted copies of the VisualWebArena and WebArena sites, reset between tasks; no live website, real account or person was contacted, and no human subjects were involved. The API-served backbones were used under their providers’ terms. Agents that act on websites can cause harm when deployed against live services; this study measures observation choices in sandboxes and releases no agent that acts outside them.

Availability. Code, task configurations, per-episode outcomes and costs, and every analysis product cited in the text (each figure and table is regenerated from them by one script) will be released with the camera-ready version.

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A Task universes and exclusions

Scored sets are the collected tasks minus protocol exclusions written as rules over task configuration: classifieds 224 of 224; VWA-reddit 203 of 205 (one task whose evaluator never checks the requested action, and one cross-site task answerable from memory); shopping 432 of 435; WebArena-reddit the 104 tasks every mode ran. Every analysis records the task set it was built on and fails if denominators differ.

B The six observation modes

Four modes carry no page screenshot and cross two factors: the text payload (the accessibility tree, or the flattened list of set-of-marks elements) and the prompt family (DOM or SoM). DOM is tree + DOM prompt; P-prompt is tree + SoM prompt; P-text is marks + DOM prompt; P-SoM is marks + SoM prompt. SoM adds the annotated screenshot to P-SoM; Vision is the raw screenshot with an empty text payload. A task’s own reference image, when it has one, is given in every mode.

C The intent rule

A task is flagged if its intent matches

```
\b(image|picture|photo|screenshot)\b|
\bcolou?r of\b|
\bhow many\b[^\.]{0,40}\bin(?:the|this)\b
```

(case-insensitive) and the task carries no reference image. It flags 71 of 224 classifieds tasks, 63 of 203 VWA-reddit tasks, 56 of 432 shopping tasks and 5 of 104 WebArena-reddit tasks. It was written while auditing reddit failures; classifieds is out-of-sample.

D Harness defects and how cells were filtered

- **GPT-5.6 Vision:** the run used a broken coordinate contract and is excluded.
- **GPT-5.6 step budget:** when the model emitted several valid actions, the parser executed one but booked the step as a free wait, so each GPT-5.6 condition took 50–90 actions beyond

the 30-step budget (2–24 for B2, none for B0 and B1).

- **Shopping search box:** typing into a search box that already holds a query submitted old + new text (“chair” + “chair with wheels” → “chairchair with wheels”) in 43% of such submissions, far more often in text modes than in Vision. A detector reproducing the defect’s per-mode counts marks affected episodes; the *clean* shopping cells drop every task affected in any mode (B0 keeps 368 of 432 tasks, B1 192).
- **Shopping state:** the category grid order differs between runs (42 order-sensitive tasks dropped), and the wishlist is not reset between tasks (4 tasks dropped).

E Six-mode results

Over all six modes, the no-interaction null of §5 is rejected in all three fully rerun cells ($p = 0.0005$, 0.017, 0.0005; Holm), with single-run reliability 0.29, 0.14 and 0.46 and three, six and two runs per task needed to reach 0.5. Across the 24 registered rerun pairs, a rerun changes between 0 and 14% of task outcomes, depending on arm and cell.

F Template and rerun robustness

Averaged over cells, the deployable excess over the frontier with folds grouped by task template is -1.60 (per-arm heads) and -0.84 (triage) points over six modes, and -0.64 and $+0.29$ over three. Fitting on one run and scoring on its rerun, averaged over the three fully rerun cells, gives -0.90 and -1.60 over six modes and -0.60 and -0.56 over three. Over six modes, every per-cell template-bootstrap interval (90%) includes zero; the three-mode readings are point estimates.

G Per-cell results

Table 2 lists the success rate and cost behind Figures 1 and 4. In shopping the fused mode is cheaper than text: its marked-element list is shorter than the accessibility tree of a product-dense page.

H Estimators

Frontier. For a cell with arms m of mean cost c_m and success rate s_m , $F(c)$ is the upper concave envelope of $\{(c_m, s_m)\}$, cut at the highest-success arm (constant beyond it) and equal to the cheapest arm’s success below the cheapest cost. Every point

cell	n	DOM		SoM		Vision	
		SR	cost	SR	cost	SR	cost
cls B0	224	17.4	6.96	27.2	7.24	25.0	6.48
cls B1	224	6.2	5.95	14.3	6.03	12.5	4.32
cls B2	224	1.3	7.68	2.2	9.08	2.2	7.07
VWA-red B0	203	14.3	10.15	14.8	11.05	7.4	9.81
VWA-red B1	203	5.9	7.33	7.4	8.00	2.5	5.24
VWA-red B2	203	3.9	9.48	1.0	11.16	2.0	6.83
WA-red B0	104	26.9	7.53	22.1	9.11	19.2	8.64
WA-red B1	104	16.3	6.58	13.5	7.94	9.6	4.47
cls B5	224	23.7	13.74	37.1	16.71	—	—
shop B0	368	12.8	12.25	16.8	9.87	16.6	7.75
shop B1	192	5.7	10.60	9.4	7.75	6.8	4.69

Table 2: Success rate (%) and mean billed cost per task (US cents) of the three deployment arms. B0 and GPT-5.6 (B5) costs are API invoices; B1 and B2 are token-priced estimates for local serving, comparable within a cell only. Shopping rows are the clean cells (Appendix D); GPT-5.6 Vision is excluded. Sources: routing_three_arm, routing_extension_cells.

of F is reached by sending a random share of tasks to each of two arms. A policy with mean cost c and success s has excess $s - F(c)$.

Router curves. Features are standardised within each training fold and fed to an L2-regularised logistic regression ($C = 1$); predictions are out-of-fold over five folds of distinct tasks. *Per-arm heads:* for each threshold $\tau \in \{0, 0.05, \dots, 1\}$ a task goes to the cheapest arm (by training-fold mean cost) whose predicted success is at least τ , and to the training fold’s best arm if none is. *Triage:* one model predicts whether any arm solves the task; for each quantile $q \in \{0, 0.05, \dots, 1\}$ of its score, tasks below the quantile go to the cheapest arm and the rest to the best arm.

Pooled gain. Within a cell, cost is normalised as $u = (c - c_{\min}) / (c_{\max} - c_{\min})$ over the fixed arms, which makes cells with different cost units comparable without pooling their costs. The gain at u is the envelope of the fixed arms together with the curve’s points minus the envelope of the fixed arms alone, on 101 grid points; the pooled curve is the equal-weight mean over cells and its statistic is the maximum over u . The null permutes each task’s outcomes and costs against its features and refits the whole curve ($B = 1000$; $p = (k + 1) / (B + 1)$; for the pooled statistic each null draw is averaged over cells before the maximum).

Deployable policy. In each outer fold the threshold (or quantile) is the one with the largest excess over the *training* tasks’ own frontier, using

in-sample predictions on those tasks; it is then applied to the held-out fold. The five folds’ decisions form one policy over all tasks, scored against the full-sample frontier. Intervals resample tasks and rerun the whole pipeline ($B = 1000$, 5th and 95th percentiles); pooled intervals average the cells draw by draw.

No-interaction null and decision study. For the task $\times$ mode $\times$ run array with $r = 2$ runs, the two-way random-effects ANOVA gives $\hat{\sigma}_{\text{err}}^2 = \text{MS}_{\text{err}}$ and $\hat{\sigma}_{\text{int}}^2 = (\text{MS}_{\text{int}} - \text{MS}_{\text{err}}) / r$. The null matrices are drawn with curveball trades over the task-by-(mode, run) 0/1 matrix, which preserve every row and column total (four chains, each with 50 trades per row of burn-in and 5 per row between kept draws, 2000 draws in all; for the six-mode analysis a second sampler with four-fold thinning checks mixing). The reliability of a label averaged over k runs is $\hat{\sigma}_{\text{int}}^2 / (\hat{\sigma}_{\text{int}}^2 + \hat{\sigma}_{\text{err}}^2 / k)$, and the runs needed for 0.5 are $\lceil \hat{\sigma}_{\text{err}}^2 / \hat{\sigma}_{\text{int}}^2 \rceil$.